

Quantum Gravitation

Problem Set 11

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Due 2026-11-15 23:59

Topic

Bounds on Medium Quanta

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Bounds on Medium Quanta. State all assumptions and conventions.

$$q = (d|s)/\sqrt{(s|s)}$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$p(m_g, g|D) \propto L_{prop} L_{scat} L_{therm} p_0$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$m_g^{95} < m_{32}$$

Show enough intermediate work that another student can reproduce the calculation.